

MAXIMAL COVERING LOCATION PROBLEMS FOR HEALTHCARE SYSTEMS: A SYSTEMATIC REVIEW

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ABSTRACT. Finding suitable locations to expand the coverage of healthcare systems is a significant challenge, especially when different types of facilities provide different types of services to the population. In this study, we report a systematic review of the maximum coverage facility location problem (MCLP), highlighting its application to the location of healthcare facilities. This study focuses on various areas within the healthcare sector, covering emergency services such as ambulances and fire stations, as well as hierarchical services including emergency departments, specialized clinics or hospitals, among others. It also examines humanitarian aid operations and the implementation of drones or electric vertical take-off and landing (eVTOL) vehicles in medical care. We present the main modeling assumptions, solution methods and applications that have been reported in the literature, especially in the location of hierarchical facilities and in the context of the COVID-19 pandemic. We also highlight research opportunities and future perspectives. MCLPs still represent a challenge for the research community, especially those related to hierarchical healthcare systems, as can be seen from the substantial number of work in recent years.

Keywords: Maximal covering location problem (MCLP), healthcare, hierarchical location, systematic review, COVID-19.

1 INTRODUCTION

Facility location problems (FLP) involve deciding where to site a specific service according to a specific objective. This field of research dates back to 1869, but significant advances have been made only after 1960 (Labbe, 1998). Determining where to locate a facility, such as a warehouse,

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plant, hospital, or school, is essential for both the public and private sectors. Revelle (1997) noted that the perspectives in this field are not only for competitive industry and commerce, but also for the public sector. This is a fundamental decision because a poor location could damage the quality of services (Daskin & Dean, 2005).

The literature on FLP classifies problems as continuous, network, or discrete, based on their location space (Labbe, 1998). The discrete class is further divided into covering-based, median-based, and other problems. We focus on coverage-based, since they are handy for helping emergency services make location decisions (Ahmadi-Javid et al., 2017) and dealing with coverage requests and limited resources, such as in public services (Farahani et al., 2012). The objective of this type of problem is to determine the optimal location while maintaining a limited distance or time between the client and the server. This is known as the covering distance.

There are three types of covering-based problems: set-covering problems, maximal-covering problems, and p-center problems, as stated by Ahmadi-Javid et al. (2017). For a more detailed classification, see Daskin & Dean (2005); Ahmadi-Javid et al. (2017).

The MCLP is suitable for healthcare services. Given the limited resources and growing demand worldwide, particularly during atypical situations such as the COVID-19 pandemic, it is crucial to select the appropriate location for healthcare services.

For healthcare facility location problems, it is more critical to choose the best site because mistakes could increase morbidity or mortality (Ahmadi-Javid et al., 2017; Corden et al., 2021). Moreover, making decisions in such areas, where budgets are often limited and high demands are met equitably, can be considerably challenging.

Besides saving lives, healthcare services encompass everything from fundamental care to advanced procedures such as surgery. Consequently, they are inherently structured in a hierarchy, with different facilities offering various degrees of service complexity.

Mathematical models, heuristics, and simulations can be useful tools in determining the most appropriate location for healthcare and hierarchical services. Recent studies such as Jang & Lee (2020) have examined the healthcare and hierarchical location of neonatal intensive care units in Korea.

To our knowledge, no prior research has conducted a systematic review of maximal covering location problems (MCLP) focusing on the location of healthcare facilities. Reviewing articles was simpler because the number of publications was smaller (Siddaway et al., 2018). Systematic reviews have become a common practice in reducing bias by synthesizing the large number of available articles. According to Moher et al. (2009), this methodology has become increasingly important in the field of healthcare.

According to Siddaway et al. (2018), a systematic review is a reproducible method that entails a comprehensive search of all articles on a specific topic. This tool aims to address a research question by showcasing the breadth, characteristics, and quality of the articles.

In this context, we aim to fill this gap by conducting a systematic literature review on MCLP and its application to the location of healthcare-related facilities. The range of services includes ambulances, clinics, hospitals, emergency departments, and additional healthcare support services, including drone deployment. This review also highlights hierarchical and Covid-related applications.

2 MATERIALS AND METHODS

This review explores the field of MCLP and presents the latest research with a specific focus on a healthcare approach. Consequently, we comprehensively reviewed research papers on solution methods, the number of service levels considered, and the connection to the COVID-19 context. This review is guided by three primary questions.

1. What are the different models and approaches that have been used for healthcare facility location problems?
2. How relevant is the use of hierarchical models for the decision-making of the location of healthcare facilities?
3. To what extent has the COVID-19 pandemic influenced research and development of healthcare facility location strategies?

By conducting this systematic review, we can provide specific and unbiased answers to these questions using a predefined protocol (Denyer & Tranfield, 2009).

We chose Scopus[®] and Web of Science[®] as our databases because they cover a wide range of international journals. These bases make it possible to download the main information of the articles with an interface for bibliometric analysis software, such as VOSviewer and Start.

The selected search string was ("Facility Location" AND "cover*"). From the total of papers obtained from the search strings in the databases (2797), we selected those from journals published in English (Fig. 1). Specifically, we selected articles that addressed MCLP after reading the abstract and excluding duplicates (560). We then analyzed the main body and analyzed the methods and analyses of 203 articles with different applications of MCLP. Among these articles, we identified 63 that addressed the following topics: healthcare, health transport, humanitarian relief, and COVID-19.

We also included review articles to assess the state-of-the-art of MCLP in the health field, resulting in a total of 74 articles for full reading. This final set of articles represents approximately 2,6% of the total articles identified in the initial search (Fig. 1).

By carefully selecting and analyzing these articles, we aim to provide a comprehensive understanding of the relationship between healthcare facility location and the MCLP, including the hierarchical systems and the location of COVID-19 services.

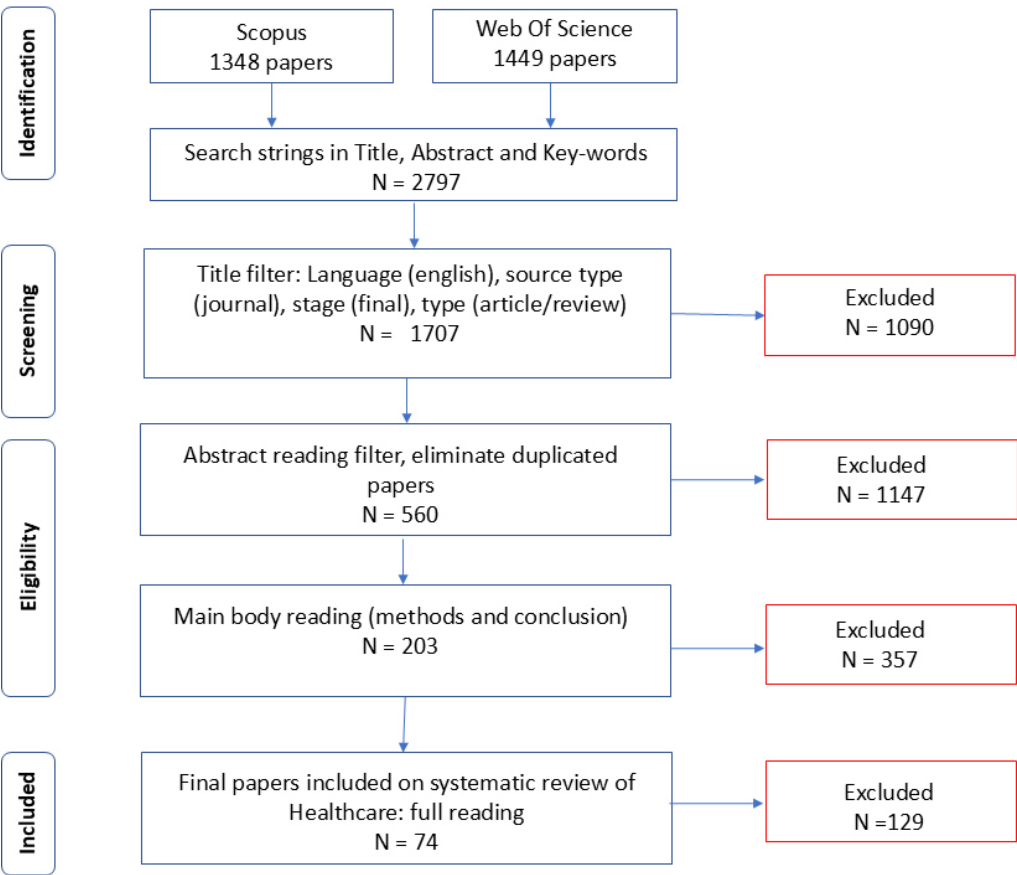


Figure 1 – Flow of information through the systematic review phases. Modified from Moher et al. (2009).

The methodology described in Table 1 was selected to reduce bias and ensure a comprehensive review of the relevant literature (Denyer & Tranfield, 2009). The methodology has three main phases: planning, execution, and systematic review. To facilitate the implementation of this methodology, we use the Start tool (Fabbri et al., 2016) to compile information and follow the necessary steps.

Before presenting the results of our systematic review of MCLP in health, we present the general characteristics of research in the field of cover facility location.

3 SUMMARY OF THE CHARACTERISTICS OF MCLP PUBLICATIONS

Bibliometric analysis helps in the study and understanding of large volumes of scientific data, such as those extracted in a systematic review (Donthu et al., 2021). We initially performed a bibliometric analysis of the search “Facility Location” and cover*, to understand the state of research on facility location.

Table 1 – Systematic review protocol.

Phase	Activities	Review Steps
Planning	Objective	Survey relation between MCLP and healthcare facility location, including hierarchy and COVID-19 applications
	Main Questions	(i) What are the different models and approaches that have been utilized for healthcare MCLP?; (ii) How relevant is the use of hierarchical models for healthcare facility location decision-making?; (iii) To what extent has the COVID-19 pandemic influenced the research and development of health facility location strategies?
	Keywords	“Facility Location” and cover*
	Source List	Scopus and Web of Science
Execution	Collection and analysis	Deleting duplicated papers
		Analyze title, keyword and abstract
		Full reading
Systematic Review	Summarize the papers selected in the previous step	Highlight the findings in accordance with the objective and questions from systematic review planning.

The bibliometric analysis of the search in the Web of Science database included 862 articles in English from 1976 to July 2024 (Fig. 2). This paper provides a bibliometric analysis exclusively of articles from the Web of Science database because VOSViewer software performs analyses on each database independently. Nonetheless, the details from Scopus closely resemble those discussed below. We selected the 100 terms cited more than 20 times in the articles. This figure shows the groups and frequency of the most commonly used terms and their connections.

We can see that although the research includes all articles on facility location in different contexts, health terms are relevant, such as patient, ambulance, emergency medical service (EMS), hospital, and even COVID. Although a recent term, COVID-19 is one of the most cited terms in this field, reflecting the significant impact of the pandemic on research.

Regarding the resolution of the problem, we verified the development of formulations and mathematical models. Furthermore, the most commonly cited solution methods were heuristics, meta-heuristics, graphs, approximation algorithm, nondominated sorting genetic algorithm (NSGA-II) and local search. In addition to analyzing the most frequent terms, we analyzed the most frequent authors (Fig. 3).

The field has three main research groups (Fig. 3), whose main authors are Oded Berman, Zvi Drezner and Allan T Murray. The latter stand out with 16 articles each, followed by Richard L. with 14. Reza Zanjirani Farahani and Murray are among the most cited authors, with 1541 and 500 citations, respectively.

Furthermore, it is worth highlighting the work of Reza Zanjirani Farahani, who, although apparently not belonging to any research group, was responsible for eight publications between 2007 and 2019, including hierarchical location work (Farahani et al., 2014) and review work (Farahani et al., 2012).

In general, this bibliometric study provides a useful overview on the MCLP and can be used to identify literature gaps or potential areas for further investigation, such as hierarchical location that is not among the most cited terms. However, it is limited by the scope of the database and search terms used and may not represent the entire research field.

After this initial analysis, we merged the information from the two databases in the *Start* software, removed duplicate articles, and further filtered the results according to inclusion criteria (MCLP, healthcare and COVID-19).

Analyzing all 203 MCLP articles found in our search, we present a summary of the characteristics. The articles were published between 1976 and July 2024 in 121 different journals, with Computers and Operations Research being the most responsible for publications in the area (Fig. 4).

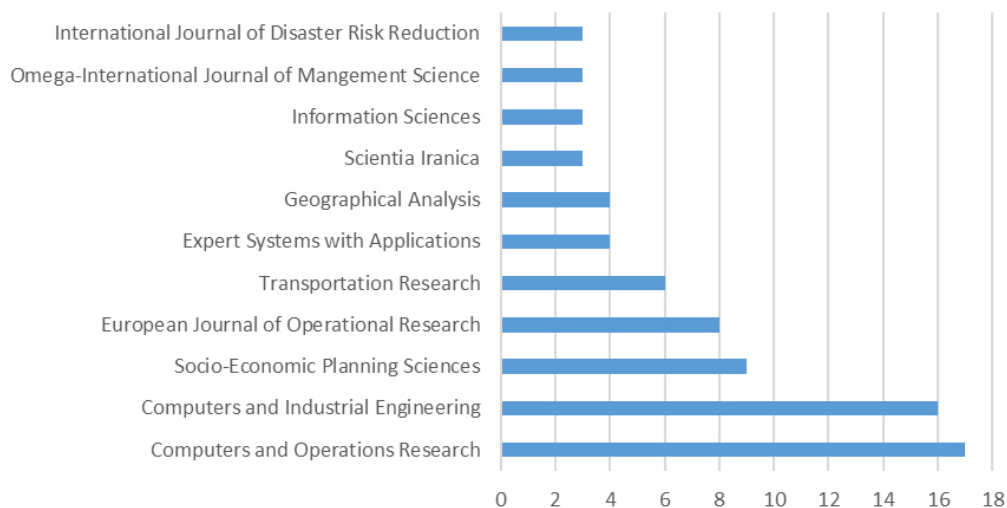


Figure 4 – Number of papers by journals.

More than 30% of the articles in the MCLP field were published in seven journals (Fig. 4). In addition, there are another 82 journals with only one publication and 36 with two publications about MCLP.

Regarding the period of publication, we observed an increasing trend in publications related to MCLP in recent years, as shown in Fig. 5. In fact, the number of publications in the last four years was approximately 70% higher than in the previous period. Taking into account the articles from the first half of 2024, it already represents 15% of the publications of the previous four years. These analyses suggest that MCLP is an active area of research and is gaining importance in several fields, including health.

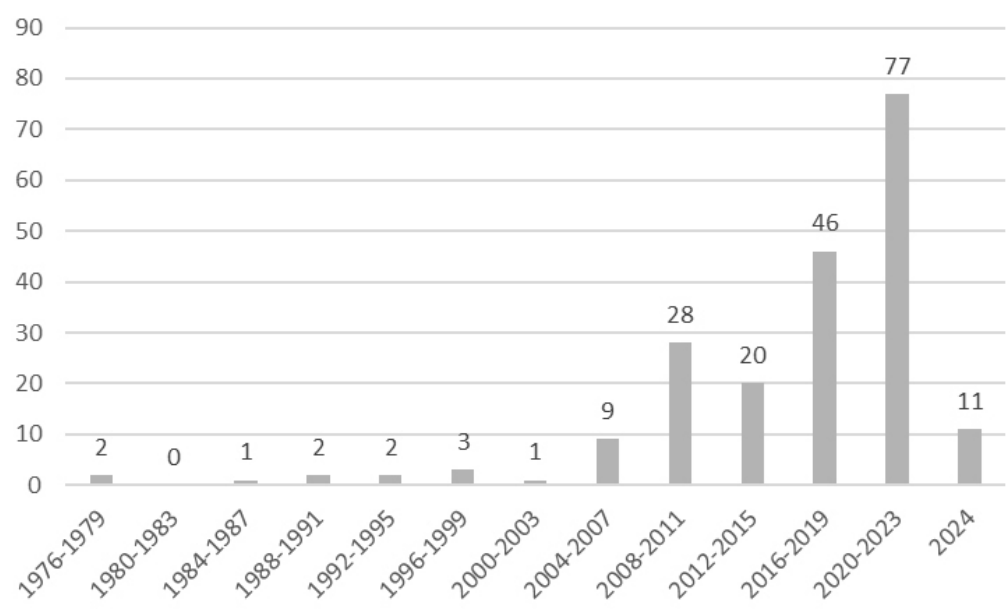


Figure 5 – Histogram of MCLP publications since 1976.

Corroborating with ReVelle et al. (1977), the utility of the MCLP model is wide-ranging, having been applied in various fields, including the location of waste and water treatment centers (Stallkamp et al., 2020), reverse logistics (Shi et al., 2019; Octarina et al., 2022), and more recently, the location of facilities in the context of the COVID-19 pandemic (Leithäuser et al., 2021; Hassan et al., 2021; Risanger et al., 2021; Delgado et al., 2022; Lusiantoro et al., 2022), as well as other applications in the healthcare sector (Fig. 6).

Furthermore, it should be noted that some of the papers identified in our review, almost 25%, were classified as "general". That is, they did not have a specific application domain, but instead focused on improving or evaluating models, methods, and solution techniques. This finding highlights the importance of ongoing research on the theoretical and computational aspects of MCLP.

It should also be noted that the review articles represent 6% of the publications. Almost half of them are concerned with the topic of health location, which will be discussed in more detail in the following section 4. In addition to the reviewed articles on healthcare location, other 63 articles on MCLP deal exclusively with related topics: health transport (Basar et al., 2011; Bagherine-

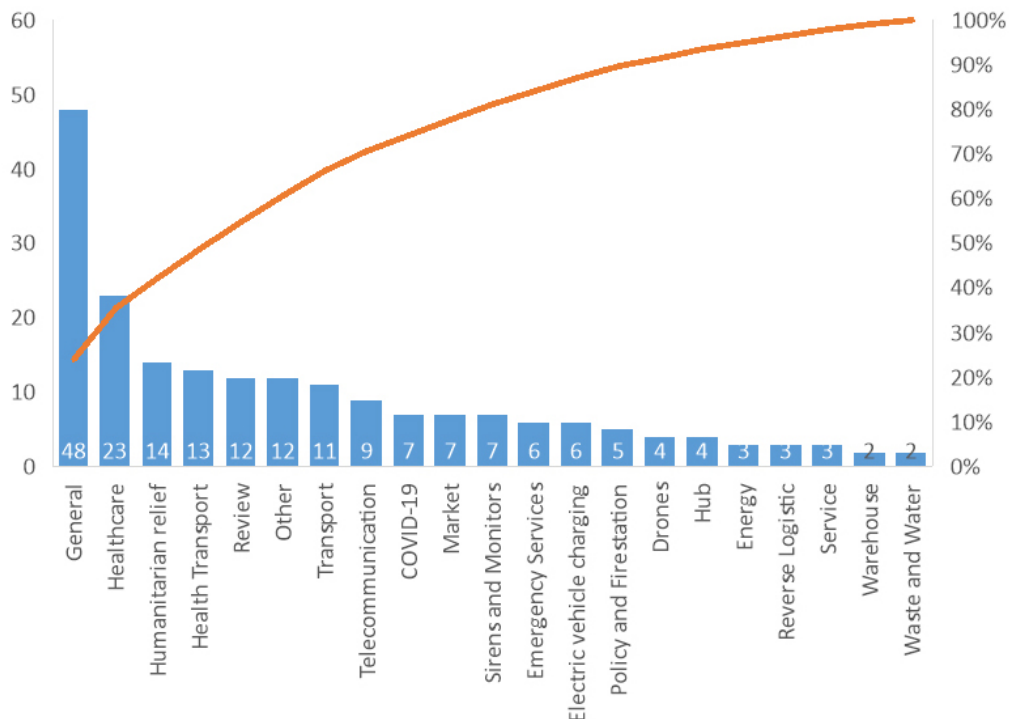


Figure 6 – Field applications of maximal covering model.

jad & Shoeib, 2018; Nilsang et al., 2019; Mohri & Haghshenas, 2021), healthcare (Pirkul & Schilling, 1989; Chen & Lai, 2022), humanitarian aid (Jia et al., 2007; Parragh et al., 2022), and COVID-19 (Risanger et al., 2021; Lusiantoro et al., 2022). These aspects highlight the relevance of the theme of the MCLP.

The use of real-world location data accounts for 44% of the publications. The remaining 126 papers on MCLP used generated data or benchmark instances. This shows the importance of using benchmarks, to compare results and develop more generic solutions, and using real data to develop solutions for specific contexts. MCLP case studies using real data are mainly concentrated in the United States and China, with 27 and 17 publications, respectively (Fig. 7).

Although Brazil has a limited number of publications in this area (Barreto et al., 2016; Ferrari et al., 2018), it ranks among the top three countries currently researching on MCLP, although considerably distant from the first two. Therefore, there is potential for further MCLP research in Brazil, particularly in healthcare, given the income disparities, demographic density, human development index, and other factors in a country of continental size.

Analyzing the solution methods used (Fig. 8), we found that 48% of the articles use exact methods: Benders decomposition (Vatsa & Jayaswal, 2016; Parragh et al., 2022), branch and bound (Singhtaun & Natsupakpong, 2017), CPLEX (Barreto et al., 2016; Liu et al., 2017), Gurobi (Al-

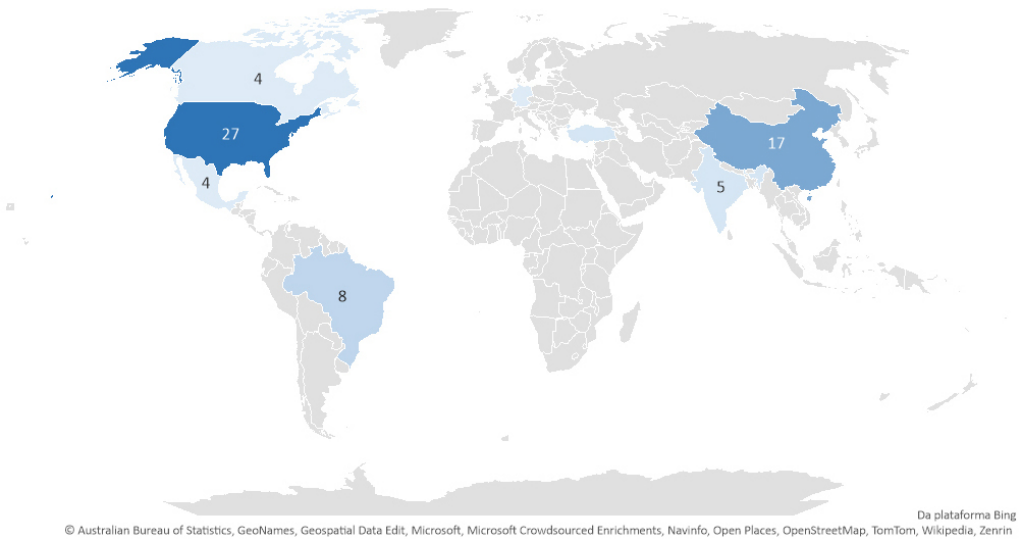


Figure 7 – Global applications of maximal covering model.

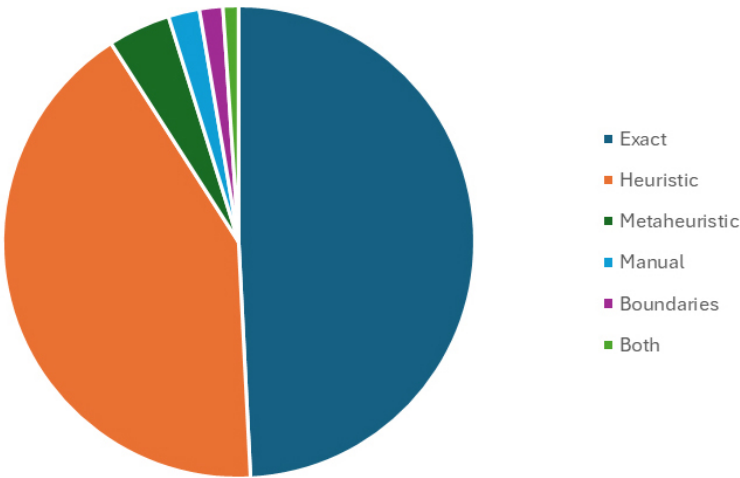


Figure 8 – Solution methods of maximal covering model.

rabiaah et al., 2022), Lindo (Bianchi & Church, 1988), and Lingo (Wang & Wang, 2010). However, for large problems, this may not be computationally viable. On the other hand, heuristics and metaheuristics, such as genetic algorithm (Bagherinejad & Shoeib, 2018), particle swarm (Elkady & Abdelsalam, 2016), greedy algorithm (Octarina et al., 2022; Al-rabiaah et al., 2022), simulated annealing (Murali et al., 2012) and tabu search (Pacheco et al., 2008), represent 41% of the used methods. In addition to those, there are yet one article using a queuing model (McLay, 2009), and another using simulation (Grot et al., 2022).

In the next section, we will summarize the papers on MCLP in the healthcare, including those related to humanitarian relief, health transport, and COVID-19 (as shown in Fig. 6). We will first present the characteristics of the model, such as methods, objectives, solutions, and instance sizes. Subsequently, in line with the proposed systematic review, we will discuss the findings related to the hierarchical location of healthcare and the location of the healthcare facility in the context of COVID-19.

4 RECENT LITERATURE REVIEW

The articles that we found using the keywords "Facility Location" and "cover*" are particularly relevant to the field of MCLP. We conducted our investigation up to July 2024 and identified several review articles related to MCLP. As mentioned in section 3, among the 12 MCLP review articles, 10 articles were found that addressed the topic of healthcare, which are summarized in Table 2.

The first review paper in 1976 studied maximal coverage in a context-free manner and for emergency medical services (EMS) (ReVelle et al., 1977). The authors highlight the wide range of applications of this model in addressing real-world needs. Since its inception in the 1970s, the model has been applied in various fields such as police, ambulances, fire locations, medical services, solid waste, production-distribution systems, sirens, telecommunications, banks, and airport locations.

Despite this versatility, all authors include the field of health service in some way. In contrast, only Farahani et al. (2014) explore the literature on hierarchical healthcare location in more detail, presenting characteristics and models, while five works only mention the approach with less explanation. This gap in the literature is also noted by Ahmadi-Javid et al. (2017), who highlight that while various health services are multi-service or multi-type, only a few papers approach facility location hierarchically.

In addition to the hierarchical model, the MCLP has other strands, as highlighted by Li et al. (2011); Farahani et al. (2012); Wang & Ma (2021): maximal coverage implicit and explicit, capacitated multi-period maximal covering Location Problem under Server Uncertainty (CMM-CLPSU), maximal available location problem (MALP), maximal expected covering location problem (MEXCLP), maximal covering location problem with location-forced" (MCLP-LF); gradual covering, backup covering problem (BACOP1 and BACOP2), bi-objective and multi-objective. Despite the various models found in the literature, the fundamental objective of the problem remains the same: maximize the population covered by a service within a limited distance R (radius) or time.

A related objective is to minimize the number of uncovered individuals, which is similar to the objective of the general p -median problem. Therefore, the MCLP can be considered as a specific case of the general p -median problem. This approach allows for the sharing of solution method-

Table 2 – Review papers for MCLP.

Author	Approach	Systematic Review? Basis	Reviewed Papers
Wang et al. (2021)	Provides a detailed discussion of problems related to emergency services covering-based, including the methods, models, and applications used to address them.	No	87
Farahani et al. (2019)	Focuses on the location of urban third-sector services, which involves making decisions regarding routing, allocation, and transportation based on facility location.	No	110
Ahmadi-Javid et al. (2017)	Reviews articles that focus on different types of healthcare facility location, both emergency and non-emergency.	No	150
Farahani et al. (2014)	Reviews facility location with a focus on hierarchical problems. It investigates the various solution methods and other characteristics of the reviewed papers to classify them.	Yes, Scopus	121
Basar et al. (2012)	Outlines a taxonomic scheme used to classify emergency service station location problems based on factors such as type, models, objective functions, constraints, and solutions.	No	57
Farahani et al. (2012)	Reviews facility location with a focus on covering problems.	Yes, Scopus	180
Li et al. (2011)	Provides an overview of covering problems in emergency response, with a specific focus on the techniques, models, and recent developments in this area.	No	114
Vijeyamurthy & Panneerselvam (2010)	Presents a literature review of covering problems, including a discussion on total and partial covering, maximal covering, and other related topics.	No	90
ReVelle et al. (2008)	Reviews two branches of discrete location: (1) the median and plant location models and (2) the center and covering models.	No	113
ReVelle et al. (1977)	Reviews facility location with a focus on emergency medical service (EMS) and context-free. Presents maximal covering problems context-free.	No	43

ologies, as demonstrated by Church & ReVelle (1976). The authors suggest a transformation in the distance parameter d_{ij} (Eq. 1):

$$d'_{ij} = \begin{cases} 0 & \text{if } d_{ij} \leq S, \\ 1 & \text{if } d_{ij} > S, \end{cases} \quad (1)$$

where d_{ij} is the real distance between the facility j and the demand point i , d'_{ij} is the modified distance between them and S is some desirable coverage service distance. In this way, the objective of the p -median problem is modified to minimize the unserved population. Thus, applying the adjusted distance parameter, the value attributed to any particular location setup corresponds to the population count that remains unserved within the specified distance S (Eq. 2).

$$\min \sum_{i \in I} \sum_{j \in J} (d'_{ij} x_{ij}) \quad (2)$$

In addition to using p -median-related approaches, other solution techniques have been used. Vijeyamurthy & Panneerselvam (2010) categorize these techniques for covering problems into different categories such as exact procedures, heuristics, metaheuristics, hybrid heuristics, and special techniques.

Exact procedures include branch-and-bound, dynamic programming, and mathematical models. Optimum techniques using commercial software are widely used due to computational advancement. However, Farahani et al. (2014) and Farahani et al. (2019) highlight that they failed for large instances, most likely faced in reality. Despite this, the exact solution may still be useful in testing the effectiveness of other methods (Farahani et al., 2014).

In this case, Vijeyamurthy & Panneerselvam (2010) highlight that using heuristics is inevitable to find a value close to the optimum. These techniques are based on guidelines to terminate the process if there is no improvement in the objective function.

In contrast, metaheuristics aim to obtain a global optimum for the problem, including techniques such as tabu search, simulated annealing, genetic algorithm, and others. Furthermore, it is possible to hybridize by combining two or more heuristics/metaheuristics (Vijeyamurthy & Panneerselvam, 2010).

According to Ahmadi-Javid et al. (2017), optimization software and heuristics are the most widely used solution techniques for covering location problems. In terms of heuristics, the genetic algorithm, tabu search, and simulated annealing are particularly noteworthy.

The reviewed articles emphasize the importance of facility location decisions for healthcare services and how they affect population access. However, the relationship between healthcare location and hierarchical systems has not been thoroughly explored, even though healthcare systems are naturally hierarchical. Only one article focuses on this area, but does not explore the topic of healthcare. To fill this gap, the following section describes the characteristics of MCLP as applied to healthcare services, highlighting hierarchical systems and the context of COVID-19.

4.1 Analysis of the literature

The location of health services is critical since it directly affects the quality, availability, and accessibility of these services to the population, being, therefore, one of the reasons that about 30% of the MCLP publications are related to health services since the first publication in 1976.

The MCLP model is an effective approach in situations where resources are scarce and full coverage is not always viable (Church & ReVelle, 1976). The main objective of this approach is to identify the optimal locations for a limited number of facilities so that the coverage of the population is maximized while being constrained by a limiting distance referred to as “covered”. The MCLP, introduced by Church & ReVelle (1974), can be stated as follows:

Sets

$i \in I$ - set of demand nodes

$j \in J$ - set of potential facilities locations

N_i - collection of potential sites that can serve a demand point $i \in I$ within distance S ,
 $N_i = \{j \in J : d_{ij} \leq S\}$

Input parameters

S - Distance limit between demand point and facilities

d_{ij} - Distance between demand $i \in I$ and location point $j \in J$

a_i - Population served at point $i \in I$

p - number of facilities to be located

Decision variables

$x_j \in \{0, 1\}$ - equal to 1 if facility $j \in J$ is installed; 0, otherwise.

$\omega_i \in \{0, 1\}$ - equal to 1 if the population of $i \in I$ is covered by some installed facility within a coverage distance of S ; 0, otherwise.

$$\max \sum_{i \in I} a_i \omega_i \quad (3)$$

$$\text{s.t.: } \sum_{j \in J} x_j = p \quad (4)$$

$$\omega_i \leq \sum_{j \in N_i} x_j \quad \forall i \in I \quad (5)$$

$$x_j, \omega_i \in \{0, 1\} \quad \forall j \in J \quad (6)$$

The objective function (3) maximizes the covered demand. In contrast, Church & ReVelle (1976) assume that this objective function is equivalent to minimizing the uncovered demand, as seen in Chen & Lai (2022). Constraint (4) installs exactly p facilities, while constraints (5) allow a demand ω_i to be covered only if at least one facility x_j is located within the distance S , that is, there is at least one installed facility in the neighborhood of $i \in I$ or N_i .

The standard MCLP model has undergone numerous improvements and adaptations, making it applicable to different research fields. Table 4 summarizes all references in the MCLP literature on health services and their characteristics. For a better understanding of the summary, Table 3 provides a list of abbreviations.

Table 3 – Abbreviations from summary papers.

Abbreviation	Meaning
C	Capacitated
EDA	Estimation of distribution algorithm
EMS	Emergency medical service
GA	Genetic algorithm
GP	Goal programming
IMOSA	Improved multi-objective simulated annealing
LB	Local Branching
LINDO	Linear interactive discrete optimizer
LR	Lagrangian relaxation
NSGA-II	Non-dominated sorting genetic glgorithm II
PSO	Particle swarm optimization
SHCS	Second-level health care services
SPEA2	Strength pareto evolutionary algorithm 2
TS	Tabu search
UC	Uncapacitated
VNS	Variable neighborhood search

Table 4 summarizes the characteristics of the 56 MCLP articles on healthcare. We briefly present the characteristics of the models, such as deterministic, stochastic, capacitated, hierarchical, and the number of objectives and periods. We also present the healthcare domain in which the model was used and the solution technique. Finally, we also highlight whether the model used is an adaptation of the MCLP.

Table 4 – An overview of MCLP for health services.

Paper	Objectives	Model Features					Field Application	Solution Technique
		Deterministic	Stochastic/ Fuzzy	Capacity	Hierarchical	Multiperiod		
de Campos et al. (2024)	$\zeta=3$	✓	-	C	-	-	Mamography	SPEA2
Hosseini et al. (2024)	2	✓	-	C	-	-	Elderly care facilities	ϵ -constraint method
Mendoza-Gómez & Ríos-Mercado (2024)	1	✓	-	UC	-	-	SHCS	CPLEX
Espejo-Díaz et al. (2023)	1	✓	-	UC	-	-	EMS	CPLEX
Gunnarsson et al. (2023)	2	✓	-	UC	-	-	Air ambulance	Gurobi
Jafarzadeh-Ghouschi et al. (2023)	$\zeta=3$	-	Fuzzy	C	-	-	Humanitarian relief	GP
Kumar (2023)	1	✓	-	UC/C	-	-	Primary health care	LINGO and CPLEX
Labita & Namoco (2023)	1	✓	-	UC	-	-	Fire station	Python PuLP
Li et al. (2023)	1	✓	-	C	-	-	Mobile health clinics	CPLEX
Vicencio-Medina et al. (2023)	1	✓	-	UC	-	-	COVID-19	EDA
Cabanilla et al. (2022)	2	✓	-	UC	-	-	COVID-19	Genetic Algorithm
Al-rabiaah et al. (2022)	1	-	Dynamic	C	-	-	Drones in healthcare	Maximum coverage greedy randomized heuristic (MCGRH)
Chen & Lai (2022)	$\zeta=3$	✓	-	UC	-	-	EMS in rural areas	IMOSA
Elorza et al. (2022)	1	✓	-	C	-	-	Primary health care	CPLEX
Goentzel et al. (2022)	1	✓	-	C	-	-	COVID-19	Solver
Grot et al. (2022)	1	-	Stochastic	C	-	-	Ambulance	Gurobi
Hashemi et al. (2022)	1	✓	-	C	-	✓	Ambulance	Genetic Algorithm
Mendoza-Gómez & Ríos-Mercado (2022)	2	✓	-	UC/C	-	-	Primary health care	CPLEX and ϵ -constraint method
Parragh et al. (2022)	2	-	Stochastic	C	-	-	Humanitarian relief	Branch-and-Benders-cut
Wajid & Nezamuddin (2022a)	1	-	Robust	C	-	✓	Ambulance	Gurobi
Wajid & Nezamuddin (2022b)	1	-	Robust	C	Two-level	✓	Ambulance	VNS and LB
Jagtenberg et al. (2021)	$\zeta=3$	✓	-	UC	-	-	Air ambulance	Gurobi

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Paper	Objectives	Model Features					Field Application	Solution Technique
		Deterministic	Stochastic/ Fuzzy	Capacity	Hierarchical	Multiperiod		
Mohri & Haghshenas (2021)	1	✓	-	UC	-	-	Ambulance	CPLEX
Vatsa & Jayaswal (2021)	1	-	Robust	C	-	-	Healthcare facilities	Benders decomposition
Alizadeh et al. (2020)	1	✓	-	C	-	-	Humanitarian relief	CPLEX
Jang & Lee (2020)	1	-	Stochastic	C	Two-level	-	Healthcare facilities	CPLEX
Maharjan & Hanaoka (2020)	1	✓	-	C	-	-	Humanitarian relief	LINGO
Hallak et al. (2019)	$\zeta=3$	✓	-	C	-	-	Humanitarian relief	Solver
Jang (2019)	1	-	Queuing theory	C	Two-level	✓	Healthcare facilities	CPLEX
Nilsang et al. (2019)	1	-	Dynamic	UC	-	✓	Ambulance	Solver
Rizeei et al. (2019)	1	-	Robust	C	-	✓	Emergency services	GIS and artificial neural network
Zhang et al. (2019)	1	✓	-	UC	-	-	Air-ground ambulance	GAMS/OSICPLEX
Bagherinejad & Shoeib (2018)	1	-	Dynamic	C	-	✓	Ambulance	GA and Bee Colony
Ferrari et al. (2018)	$\zeta=3$	✓	-	C	-	-	Ambulance	CPLEX
Mousazadeh et al. (2018)	$\zeta=3$	✓	Robust	C	Three-level	✓	Healthcare facilities	GAMS/CPLEX
Roislien et al. (2018)	1	✓	-	UC	-	-	Ambulance	CPLEX
Burkart et al. (2017)	2	✓	-	C	-	-	Humanitarian relief	ε -constraint method, complete enumeration, NSGA-II
Liu et al. (2017)	1	✓	-	UC	-	-	Ambulance linked to hospital	CPLEX
Meskarian et al. (2017)	1	✓	-	UC	-	-	Healthcare facilities	Heuristics, metaheuristics and exact

Continued on next page.

Table 4 – continued from previous page

Paper	Objectives	Model Features					Field Application	Solution Technique
		Deterministic	Stochastic/ Fuzzy	Capacity	Hierarchical	Multiperiod		
Singhtaun & Natsupakpong (2017)	1	✓	-	UC	-	-	Location transportation in humanitarian relief	Parallel two-phase branch and bound
Barreto et al. (2016)	1	-	Queuing theory	UC	-	-	Ambulance	Hypercube model with CPLEX
Elkady & Abdelsalam (2016)	2	✓	-	UC	-	-	Healthcare facilities	PSO and NSGA-II
Luo & Yang (2016)	$i=3$	✓	-	UC	-	-	Humanitarian relief	Ant colony
Mohammadi et al. (2016)	$i=3$	-	Stochastic	C	-	-	Humanitarian relief	MOPSO
Vatsa & Jayaswal (2016)	1	-	Robust	UC	-	✓	Healthcare facilities	Benders decomposition
Zhao & Chen (2015)	$i=3$	✓	-	UC	-	-	Humanitarian relief	Multi-objective evolutionary algorithms and NSGA-II
Farahani et al. (2014)	1	-	Probabilistic	UC	Three-level	-	Healthcare facilities	Hybrid artificial bee colony (HABC)
Murali et al. (2012)	1	✓	-	C	-	-	Humanitarian relief	Simulated annealing
Basar et al. (2011)	1	✓	-	UC	-	✓	Ambulance	TS
Gu et al. (2010)	2	✓	-	UC	-	-	Healthcare facilities	Interchange Algorithm
McLay (2009)	1	-	Hypercube queuing model	UC	Two-level	-	Ambulance	Hypercube queuing
Murawski & Church (2009)	1	✓	-	UC	-	-	Rural health ambulance	CPLEX
Balcik & Beamon (2008)	1	-	Probabilistic	C	-	-	Humanitarian relief	CPLEX
Pacheco et al. (2008)	1	-	Probabilistic	UC	-	-	Healthcare facilities	Scatter search, TS and VNS
Jia et al. (2007)	1	✓	-	UC	-	-	Humanitarian relief	GA, a locate–allocate heuristic, and a LR
Mandell (1998)	1	-	Probabilistic	UC	Two-level	-	EMS	CPLEX
Pirkul & Schilling (1989)	1	✓	-	C	Two-level	-	EMS	LR

Continued on next page.

Table 4 – continued from previous page

Paper	Objectives	Model Features					Field Application	Solution Technique
		Deterministic	Stochastic/ Fuzzy	Capacity	Hierarchical	Multiperiod		
Bianchi & Church (1988)	1	-	Probabilistic	UC	-	-	Emergency Facilities and ambulance	LINDO

Analyzing the Table 4 it is possible to see that approximately 65% of MCLP research in healthcare is deterministic and does not add uncertainties or probabilities to the model. Jang & Lee (2020) emphasizes that designing mathematical models that consider realistic features, such as congestion, is important. Despite this, the number of papers has still been limited to 21 since 1976.

In addition, more than 60% of the articles address server capacity, and almost 70% do not address different objectives. This highlights the importance of health service capacity. However, only 19 articles address multi-objective issues, revealing a significant research gap in multi-objective MCLP problems within the health sector.

Conversely, the assessment of solution approaches reveals an equilibrium between the application of solvers, namely Gurobi and CPLEX, and heuristics like GA, VNS, and TS, with them being utilized at rates of 41% and 37% correspondingly.

The aspects of greatest divergence were in relation to the use of hierarchical models and the multi-period approach, with 6 and 10 articles each. Consequently, there is a significant opportunity for research focusing on hierarchical and multi-period models within the MCLP in healthcare.

The subsequent sections delve deeper into the methods described in the articles, elaborating on models, solution techniques, and modifications of MCLP to address various challenges of the healthcare setting. Following this summary, we outline the specifics of hierarchical location and the COVID context.

4.2 Model features

Selecting an appropriate objective function is crucial in formulating the MCLP model, as it can significantly impact the outcomes (Jagtenberg et al., 2021). The objective function can be formulated with single or multiple objectives.

Pirkul & Schilling (1989) were the first to expand the standard MCLP in two dimensions: capacity and multilevel allocation, such as backup covering. In emergency services, especially ambulances, demand cannot be queued, so it is necessary to allocate both a primary service and a backup in case the primary service is busy when called.

The model follows the basic MCLP structure of maximizing coverage with a fixed number of facilities. However, it ensures that all demands are assigned to at least one facility. In addition, it allocates primary and backup services to different facilities up to the point of demand, considering facility capacity constraints (Pirkul & Schilling, 1989).

The authors assign equal weight to primary and backup coverage in the model (Eq. 7). However, they also suggest the possibility of a multi-objective model that can account for tradeoffs between these two services.

$$\max \sum_{i \in I} \sum_{j \in J} (c_{ij}^p a_i^p x_{ij}^p + c_{ij}^b a_i^b x_{ij}^b) \quad (7)$$

where, (i) I , is the set of all demand points (ii) J is the set of all potential facility sites (iii) x_{ij}^p, x_{ij}^b is 1 if a facility provides primary (p) or backup service (b), 0 otherwise (iv) a_i^p, a_i^b is the demand for primary (p) or backup service (b) (v) $c_{ij}^p = 1$ if $d_{ij} \leq S^p$ or 0, otherwise (vi) $c_{ij}^b = 1$ if $d_{ij} \leq S^b$ or 0, otherwise (vii) S^p, S^b the maximum service distance or time for primary (p), backup (b) (viii) d_{ij} travel distance or time from $j \in J$ to $i \in I$

The objective function (7) maximizes the demand assigned to each service level that is covered by a set of facilities. In this type of problem, any facility is capable of delivering merely one level of service to each individual node. This means that no facility can provide both services to the same point of demand.

Pirkul & Schilling (1989) point out a limitation in the conventional model format for backup services, where uncovered demand is not prompted to connect with proximate facilities. Indeed, a demand point that is not already covered will be allocated to any available facility, irrespective of its distance. Although this fulfills the goal of the model, it does not result in a practical allocation pattern. In addition, this model considers binary multipliers c_{ij}^p and c_{ij}^b , in which the discovery demand does not contribute to the objective function and is not necessarily attributed to any facility.

In contrast, Pirkul & Schilling (1989) employ binary multipliers, denoted c_{ij}^p and c_{ij}^b , to restrict consideration to sites that fall within the maximum distance for the primary or backup service. Another notable aspect is the gradual reduction in the contribution of uncovered demand points by a distance-dependent function, as seen in Eq. 8 - 9.

For the primary service $c_{ij}^p = 1$ if $d_{ij} \leq S^p$, or

$$c_{ij}^p = \alpha \left(1 - \frac{(d_{ij} - S^p)}{(d_{\max} - S^p)} \right) \quad (8)$$

and for the backup service $c_{ij}^b = 1$ if $d_{ij} \leq S^b$, or

$$c_{ij}^b = \alpha \left(1 - \frac{(d_{ij} - S^b)}{(d_{\max} - S^b)} \right) \quad (9)$$

where $d_{\max} = \max_{i \in I, j \in J} d_{ij}$ and $\alpha \in (0, 1)$ is a scale factor that regulates the gap between met and unmet demand. This definition assigns a uniform weight of one to all demands allocated to a facility located within the coverage distance, while demands allocated to facilities farther away receive a gradually diminishing weight (Pirkul & Schilling, 1989).

Conversely, Mandell (1998) considers the probability that each demand point will be covered within an appropriate timeframe. This is the idea behind the maximum expected coverage problem (MEXCLP) proposed by Daskin (1983). The authors add the probability of a demand being met properly in the objective function. Additionally, they adopt a hierarchical approach to address the problem at two distinct levels of emergency, basic life support units (BLS) and advanced life support (ALS).

According to Mandell (1998), BLS required less advanced care, such as cardiopulmonary resuscitation (CPR) and oxygen administration. On the other hand, advanced life support (ALS)

performs more complex procedures such as life-saving, medication administration, breathing tube insertion, and administering debriding shocks, in addition to BLS.

Jang & Lee (2020) also use the expected coverage and hierarchical characteristics, adding a capacity allocation strategy for the neonatal care system in Korea, as seen in the objective function (10). Furthermore, the authors incorporate the referral of patients from the lower to the highest level and queuing theory to address patient waiting times.

Although Jang & Lee (2020) consider only levels II and III in their approach, they classify the three levels of neonatal intensive care units (NICUs) as: (i) Basic level of maternity care and assessment of the newborn, transferring the newborn to a higher-level unit if necessary. (ii) Care for moderately ill patients but with an expected faster recovery; (iii) Specialized care for newborns with less than 28 weeks of life or who are critically ill.

The objective function is given by the sum of the expected covered rate of non-severe (NS) and severe (S) patients per unit time (t) at level II ($j \in J$) or level III ($k \in K$), and the coverage of referral patients from level II to level III.

$$\begin{aligned} \max \sum_{t \in T} \sum_{i \in I, j \in J_i} S_{ij}^{NS,t} + \sum_{t \in T} \sum_{i \in I, k \in K_i} S_{ik}^{NS,t} \\ + \sum_{t \in T} \sum_{i \in I, k \in K_i} S_{ik}^{S,t} + \sum_{t \in T} \sum_{j \in J, k \in K_j} S_{jk}^{S,t} \end{aligned} \quad (10)$$

where, (i) $S_{ij}^{NS,t}$ is the expected covered rate of non-severe patients per unit time from i to level-II NICU j ; (ii) $S_{ik}^{NS,t}$ is the expected covered rate of non-severe patients per unit time from i to level-III NICU k ; (iii) $S_{ik}^{S,t}$ is the expected covered rate of severe patients per unit time from i to level-III NICU k ; (iv) $S_{ij}^{S,t}$ is the expected covered rate of severe patients per unit time from i to level-II NICU j . To address the effect of congestion at an NICU, Jang & Lee (2020) uses the Erlang-C formula that represents the probability that a patient at an NICU experiences delays in services.

Similarly, McLay (2009) presents the MEXCLP model with two types of servers, named MEXCLP2. The key distinction highlighted by the author is that while the original MEXCLP addresses the placement of a single server type for one client type, MEXCLP2 expands this scope by incorporating two server types catering to various client types. Furthermore, the study involves three levels of client priorities. Using the Hypercube model, the correction factors and busy probabilities were calculated.

The author highlights the benefits of this approach, as it accounts for various categories of clients and the interrelatedness of EMS responses for each type of client. In MEXCLP2, the aim is to optimize the total expected coverage of Priority 1 calls within a given timeframe. A call is considered covered when an ALS or BLS medical unit reaches the location within this time limit, with priority given to dispatching ALS units first if one is accessible nearby.

Another possibility is to hybridize MCLP with other models. Bianchi & Church (1988) proposed a hybrid approach to the MEXCLP and facility location and equipment emplacement

technique (FLEET) model, called the multiple cover, one-unit FLEET problem (MOFLEET). This approach maintains the original goal of MEXCLP while simultaneously locating a fixed number of stations and allocating a specific number of ambulances.

The objective function aims to minimize the uncovered population, given the probability of a randomly selected ambulance not being available (u).

$$\min \sum_{i \in I} \sum_{k \in K} (1 - u)^{k-1} a_i y_{ik} \quad (11)$$

where, (i) a_i is the population to be served at demand node $i \in I$; (ii) u is the probability of a randomly selected ambulance being unavailable; (iii) $y_{ik} = 1$ if demand node $i \in I$ is not covered by k ambulances; 0 otherwise.

Farahani et al. (2014) used a similar objective function and proposed the reliability of hierarchical maximal covering (RHMCLP) by considering the probability of disruption due to uncertainty (natural or man-made). The authors focus on the issue of reliability and disruptive facilities for three-level healthcare systems.

The authors developed a model considering three levels of service, and each facility performs only one of the levels, considering the possibility of failure.

Building on MEXCLP, Grot et al. (2022) proposed two advancements, incorporating coverage restrictions and probabilistic capacities. The capacitated maximum expected covering location problem (CMEXCLP) established new chance constraints to limit the business of each site. Meanwhile, the level maximum expected covering location problem (LMEXCLP) introduced probabilistic coverage into the framework.

Therefore, another important consideration when formulating the MCLP model is the uncertainty of the parameters, such as travel time, demand, and capacity. Deterministic formulations, which account for nearly 60% of the models, disregard such uncertainties. However, these models may not always achieve optimal results in real-world scenarios due to factors such as rapidly changing conditions, lack of data, and noise, which can compromise their effectiveness (Rahmaniani et al., 2014).

Researchers have used approaches such as queuing theory (Barreto et al., 2016; Jang, 2019), stochastic (Mohammadi et al., 2016; Jang & Lee, 2020; Parragh et al., 2022), Fuzzy (Jafarzadeh-Ghouschi et al., 2023), probabilistic (Bianchi & Church, 1988; Mandell, 1998; Pacheco et al., 2008; Balcik & Beamon, 2008; Farahani et al., 2014) models to address the challenges of uncertainty. These methods consider the uncertainties and provide more realistic solutions to MCLP problems.

These features result in extensions of the original MCLP. For example, Murawski & Church (2009) considers uncertainties about road conditions throughout the year, an aspect that is usually ignored in models, despite being relevant in rural areas. The authors emphasize the limited usefulness of location allocation models in improving accessibility when road conditions are ignored. Given the challenges of reconfiguring the facilities, the authors recommended improving

road conditions to maximize population coverage throughout the year. To achieve this goal, they developed the maximum coverage network improvement problem (MC-NIP).

The objective function shares similarities with Eq. 3, but it differs in that ω_i takes into account the presence of an all-season route connecting a village and the clinic within the coverage distance.

Chen & Lai (2022) also tackled the issue of roads in rural areas and underscored the lack of research carried out in this field. The authors propose a maximal covering facility location network design (MCFLND) varying assumptions:

1. covering scenarios where existing roads cannot be improved;
2. situations where existing roads can be improved; and
3. cases where existing roads can be improved while also constructing new ones.

The three objectives considered in their study were to minimize unmet demand, minimize the total time that patients took to reach the nearest facility, and minimize construction costs.

On the contrary, the limitations of health services in densely populated cities like Hong Kong are emphasized in Zhang et al. (2016), underscoring the need to address the complex multi-objective problem of location of healthcare facilities.

Another approach used is the multi-period approach, which considers coverage over a given period of time. Basar et al. (2011) addressed a multi-period ambulance location problem, to maximize the double covered population, using the multi-period backup double covering model (MPBDCM).

Vatsa & Jayaswal (2016) also address the multi-period problem introducing two novel models considering uncertainties: multi-period maximal covering location problem under server uncertainty (MMCLPSU) and multi-period maximal covering location problem with partial covering (MMCLP-P). The authors analyze the availability of doctors to primary health care units.

The objective function remains the same as in Eq. 3, only the constraints differ. A subsequent work of Vatsa & Jayaswal (2021) extended these models by including capacity constraints, resulting in the capacitated multi-period maximal covering location problem under server uncertainty (CMMCLPSU) model.

The objective functions of these methods are similar to Eq. 3, with the difference that, in this case, it maximizes the demand covered across the planning horizon (12).

$$\zeta_s^* = \max \sum_{i \in I} \sum_{j \in N_i \cup N_i^b} \sum_{t \in T} a_{ij} x_{ijts} \quad (12)$$

where, (i) a_{ij} is the proportion of demand at node i covered by facility $j \in J \cup J^b$ $a_{ij} \in [0, 1]$;
(ii) x_{ijts} is the demand at $i \in I$ assigned to facility $j \in J \cup J^b$ in period $t \in T$ and scenario $s \in S$;
(iii) ζ_s^* is the maximum demand that can be covered over the complete planning horizon in scenario s .

The authors additionally offer a linearization of the objective function while incorporating uncertainties related to servers. Refer to Vatsa & Jayaswal (2021) for further details. It should be noted that the proposed model considers the opening of new facilities $j \in J$ besides those that are already operating J^b .

Mousazadeh et al. (2018) also addressed multi-period problems by adding a hierarchy with three levels. The authors proposed a mixed-integer non-linear programming (MINLP) model with a more practical objective function, the weighted-sum maximal covering. Their emphasis was on the placement of various healthcare facilities (such as primary health centers, clinics, and hospitals), the potential closure of current facilities, facility capacity, patient distribution to active facilities, and the referral patterns among operating facilities over time.

This function aims to increase accessibility by placing facilities in densely populated regions, since each zone is weighted in the objective function in proportion to its demand. More recently, Hashemi et al. (2022) and Wajid & Nezamuddin (2022b) have also addressed the multi-period problem, to ambulance location.

The literature has shown significant interest in multi-objective models with diverse adaptations and applications to address the MCLP. For example, Gu et al. (2010) proposed a bi-objective model that considered not only covering but also efficiency, which was represented by maximization of social welfare.

Similarly, Mohammadi et al. (2016) presented a multi-objective stochastic programming model localization but in a humanitarian aid context. The objectives considered are to maximize the total expected demand covering, minimize the total expected cost, and minimize the difference in the satisfaction rate between nodes.

Parragh et al. (2022) focused their research on the bi-objective stochastic facility location problem (BOSFLP) in the context of drought disasters. The aforementioned study aimed to simultaneously maximize the expected demand covering and minimize the expenses linked with facility establishment.

Recent studies have employed the multi-objective method for positioning COVID-19 vaccination centers (Cabanilla et al., 2022) as well as for determining the location of mammography machines (de Campos et al., 2024). Several other authors also consider the multi-objective problem (Zhao & Chen, 2015; Elkady & Abdelsalam, 2016; Burkart et al., 2017; Ferrari et al., 2018; Mousazadeh et al., 2018; Mendoza-Gómez & Ríos-Mercado, 2022; Hosseini et al., 2024). In general, objectives involve demand, costs, and distances.

In their work on shelter locations in Syria, Hallak et al. (2019) introduced a mixed-integer weighted-goal programming approach. This involved considering nine different objective functions, including minimizing total costs, maximizing the number of internally displaced people, and four objectives aimed at maximizing the coverage of vulnerable populations, such as children, pregnant women, and those with chronic diseases. The selection of these objectives was

based on the specific needs of the scenario, which highlights the critical role of appropriately aligning the objective functions with the desired results.

Finally, recent technological advances have led to the creation of novel maximal covering models tailored to the drone field, such as the maximal covering facility location problem with drones (MCFLPD). Al-rabiaah et al. (2022) proposed this problem with an objective function similar to the standard MCLP. The authors introduced a greedy randomized heuristic for maximum coverage to tackle the issue. Drones have the potential to enhance healthcare services, ranging from medication delivery to providing patient care.

An important factor to be taken into account in localization challenges, especially in healthcare, besides the model, is the presence of uncertainties. During our investigation, we identified just 21 papers addressing uncertainties in their formulations, which will be detailed in the following section.

4.3 Handling uncertainty

Dealing with uncertainties and different probabilities is a great challenge when locating facilities, especially in multiperiod problems, as proposed by Vatsa & Jayaswal (2016). Most articles on the topic did not address uncertainty. Only 36% used fuzzy, probabilistic, robust, or stochastic models with various approaches, as described in this section.

A fuzzy programming approach to deal with uncertainty has been the least widely used, with only Jafarzadeh-Ghouschi et al. (2023) using it in the context of humanitarian supply chains. The authors considered supply, demand, and communication routes between affected centers and areas defined as fuzzy random parameters.

On the other hand, probabilistic approaches were widely used in early research between 1988 and 2008 (Table 4). For example, Bianchi & Church (1988) used the MEXCLP model to locate ambulances, where expected coverage is determined by dividing the daily workload by the number of available servers.

McLay (2009) extended this model to include two servers (MEXCLP2) and used the Hypercube model to estimate busy probabilities and the fraction of time to assign a call to the server.

In contrast, other studies (Balcik & Beamon, 2008; Farahani et al., 2014; Mohammadi et al., 2016) focused on addressing the likelihood of adverse scenarios in the context of humanitarian relief. Mohammadi et al. (2016) emphasized the importance of defining scenarios, assigning probabilities, and determining parameters when modeling uncertainties. Consequently, stochastic programming (SP) has proven to be a more effective tool than other approaches to address uncertainties in disaster relief planning.

An additional aspect of uncertainty addressed in the context of disasters is the potential for failure of the installation. Farahani et al. (2014) proposed a more realistic model that accounts for the likelihood that a facility is not operating properly.

One common strategy to address uncertainty in facility location issues involves the use of queuing models. Notable examples include the two-dimensional queuing model introduced by Mandell (1998) to examine server availability; the hypercube queuing model used to predict the likelihood of servers being busy and the response time to ambulance calls, as described in McLay (2009) and Barreto et al. (2016); and the Erlang-C formula which incorporates time factors and the randomness of each NICU server as discussed in Jang & Lee (2020).

In contrast, Vatsa & Jayaswal (2016) pointed out that the robust model is useful in the absence of these probabilities and proposes a regret approach (13).

$$\min_{\pi \in \Pi} \max_{s \in S} (\zeta_s^* - \zeta_s^\pi) \quad (13)$$

where, (i) ζ_s^* is the optimal coverage in scenario $s \in S$; (ii) ζ_s^π is the coverage obtained with any proposed facility opening sequence $\pi \in \Pi$; (iii) Π is the set of all possible sequences of opening facilities $j \in J$.

The regret in each scenario s is determined by computing the difference between the maximum population that can be covered ζ_s^* and the actual population covered obtained by 12. In a subsequent work, Vatsa & Jayaswal (2021) added server capability to their model.

In contrast, Mousazadeh et al. (2018) emphasized the importance of robust decision making in a hierarchical location-allocation problem, given the strategic nature of these decisions which can be difficult, if not impossible, to revise. The authors propose a robust mixed possibility-flexible programming approach that considers input data uncertainty while treating some equations as soft constraints.

In this context, Rizzei et al. (2019) utilized machine learning techniques to address uncertainty in the optimal allocation of emergency response centers (ERCs). The researchers developed an optimized multi-layer perceptron (MLP) that accounts for the dynamics of pluvial flooding (PF). They employed the robust design principles of the Taguchi approach to enhance the MLP by choosing the most precisely calibrated values for the key influential factors. More recently, Wajid & Nezamuddin (2022b) and Wajid & Nezamuddin (2022a) developed robust approaches to the location of ambulances in Delhi, India, incorporating uncertainty about travel times for reliable EMS performance.

Another approach used is dynamic models, which consider variations over time. Bagherinejad & Shoeib (2018) have developed a model for ambulance allocation that incorporates dynamic capacity constraints for facilities. Their model, known as the dynamic capacitated MCLP, takes into account uncertainties over a specified period T . Furthermore, the authors have suggested that it is possible to transform static models into dynamic ones.

In a similar effort, Nilsang et al. (2019) developed the location-forced maximal covering location problem (MCLP-LF) model that integrates social media data into its optimization framework to establish the placement of an ambulance. This model is adaptable and enhances traditional MCLP by incorporating a location-based constraint gathered from social media data. Suppose a control chart analyzing word counts from Twitter detects an anomaly. In that case, the model

imposes a restriction to necessitate a change of base for ambulances from low-demand areas to high-demand areas.

Another common approach to the location allocation of ambulances is the use of a two-stage model, as demonstrated in several studies (Barreto et al., 2016; Liu et al., 2017; Zhang et al., 2019). These authors have taken into account various factors at each stage. For example, Liu et al. (2017) have emphasized the importance of the distance or time between the ambulance and the demand point, as well as the accessibility of the hospital to provide necessary medical care.

Liu et al. (2017) have also divided hospitals and patients into two groups depending on the level of service required. However, they have not discussed the implementation of a hierarchical model since the objective function does not target hierarchical covering but rather double covering as a result of the busy fraction of servers.

However, upon analyzing the articles concerning the hierarchical approach, it becomes apparent that over 85% of them favor single-level installation locations. Furthermore, the placement of three-layer locations comprises less than 4% of the total.

To address our main research questions (ii) and (iii) in Table 1, we will perform a brief analysis of the articles exploring the hierarchical location of the COVID-19 facilities in the following section.

4.4 Hierarchical healthcare location

Narula (1984) proposed a classification specifically for hierarchical location-allocation problems. According to the author, hierarchical location problems can be classified according to type and flow.

Type refers to whether the lower-level facility offers only essential services and higher-level facilities provide more complex services, including those offered by the lowest levels, which is known as nested or successively inclusive. If each facility only provides services at its own complexity level, they are considered nonnested or successively exclusive.

In terms of flow, Narula (1984) suggests that hierarchical location problems can be categorized as integrated if there is flow from any lower level to any higher level, or discriminating if the flow is only from the lower level to the next higher level.

According to Mandell (1998), EMS facilities have a nested structure that is commonly used in the literature. However, others suggest a backup covering that is similar to hierarchical systems in their study (Pirkul & Schilling, 1989). This approach involves providing two types of service to the same demand point, but always by different facilities. It is suitable for situations where demand cannot be queued, such as ambulance services as noted in Pirkul & Schilling (1989).

The urgency of medical services typically makes it impractical to wait, leading to the common practice of treating them as a capacitated system. However, in certain situations, queueing can be deemed acceptable, allowing them to be treated as an uncapacitated system, as evidenced in Farahani et al. (2014) and Mandell (1998).

Analyzing each paper's objective function, in general, they aim to maximize the demand covering. Furthermore, they can maximize the covered demand on the planning horizon (Jang, 2019), or maximize the expected covered demand while considering the risk of disruptions for different levels of facilities (Farahani et al., 2014).

Despite the apparent similarities between the problems, there are notable differences that affect their impact and solutions: uses of queuing theory and capacitated system (Jang & Lee, 2020), (Mandell, 1998); delimited problem (neonatal location) and multi-flow (Jang & Lee, 2020); or consider uncertainty (Farahani et al., 2014). Therefore, it is important to carefully analyze the unique features and requirements of each problem to develop appropriate solutions. The size of a two-tier maximal covering location problem, as determined by the number of nodes and facilities, significantly impacts both the time required to find a solution and the quality of the solution obtained.

The majority of bi-level challenges focus on determining the optimal placement and assignment of ambulances, involving basic life support (BLS) and advanced life support (ALS), as outlined by Pirkul & Schilling (1989); Mandell (1998); McLay (2009); Wajid & Nezamuddin (2022b). In their formulation, the problem was designed to maximize expected dual coverage, distinguished by McLay (2009), who incorporates a time-bound coverage criterion.

Conversely, Jang & Lee (2020); Jang (2019) uniquely tackled the placement of healthcare facilities across two tiers within Korea's neonatal care network, as represented in the objective function (10). The authors examined the positioning of the intermediate level, catering to patients with moderate illnesses, and the advanced level, targeting premature infants, while also taking into account potential flows between these levels (multi-flow). However, it is only Jang (2019) who incorporates a multi-period approach to the problem.

The literature for two-tier maximal covering location is limited to small problems, with 55 nodes and different combinations of installations ranging from 0 to 10 for each level (Mandell, 1998), and large problems with locating more than 120 facilities in 3482 nodes (Jang & Lee, 2020; Jang, 2019), all of them using CPLEX to solve the problems.

This feature is directly related to the solution method used. For problems involving only small instances, solvers, such as GUROBI, CPLEX, or Xpress-MP, are often used as effective solutions (Mandell, 1998; Jang & Lee, 2020; Jang, 2019). However, large-size instances use lagrangian relaxation (Pirkul & Schilling, 1989), hypercube queuing (McLay, 2009), variable neighborhood search (VNS) meta-heuristic with the local branching algorithm (Wajid & Nezamuddin, 2022b). In such cases, using solvers to solve these problems would be impractical, as it would be a time-consuming and burdensome process.

Regarding capacity analysis, only Mandell (1998) and McLay (2009) are non-capacitated models. Regarding capacity analysis, only Mandell (1998) and McLay (2009) are non-capacitated models. Regarding uncertainties, we find: travel time (Wajid & Nezamuddin, 2022b); delay in queueing (Jang & Lee, 2020); availability of an ambulance (McLay, 2009; Jang, 2019) and probability that the demands at each node are adequately served (Mandell, 1998).

Regarding the three-tiered hierarchical location models, the issue is examined solely by Farahani et al. (2014); Mousazadeh et al. (2018). They segment service levels into primary health centers (PHC), regional health centers (RHCs), and district health centers (DHCs). Farahani et al. (2014) on the other hand addresses the failure probability due to disruptions, while Mousazadeh et al. (2018) deals with epistemic uncertainty due to the dynamic nature of the problem that does not always behave according to historical data.

Solving three-tier problems can be challenging. As such, Mousazadeh et al. (2018) proposed a multi-objective model that involves facility redesign over a three-year horizon. However, this model was only tested on a small-scale instance with ten nodes and was solved using the CPLEX package. For medium and large instances, using solvers may not be practical. In such cases, the use of heuristics and metaheuristics becomes an alternative solution approach (Mousazadeh et al., 2018), as addressed in previous research (Farahani et al., 2014) to solve the problem with 400 nodes and 110 facilities.

As the number of nodes and facilities increases, the number of variables and constraints also increases dramatically, making it challenging to use a solver to find the optimal solution (Chouksey et al., 2022). However, this problem can be mitigated by simplifying the problem, incorporating appropriate constraints into the model (Chouksey et al., 2022), limiting the number of locations (Vieira et al., 2019), relaxing the model or using heuristics and metaheuristics. These methods typically yield satisfactory solutions within a reasonable time frame. It is also worth highlighting recent works that address only one of the hierarchical levels for the location of healthcare facilities, such as Kumar (2023) which addresses the first level, and Mendoza-Gómez & Ríos-Mercado (2024) which addresses only the second.

Comprehensive literature on healthcare hierarchical maximal covering location analysis has several research opportunities that are worth of being addressed in the future. These gaps include exploring the incorporation of uncertainties, capacities, and additional objectives into the problem, testing the model for larger instances, and developing effective heuristics or metaheuristics to solve medium- and large-scale instances.

While various problem characteristics and approaches exist, they all emphasize the usefulness of models in assisting health managers with scenario analyses to enhance health services in real-world situations. Ultimately, the number and capacity of facilities are directly linked to efficiency and accessibility.

Although creating models that accurately reflect reality may be difficult, they can prove extremely valuable in times of disruption, such as during the COVID-19 pandemic. This infectious disease affected more than 500 million people and claimed the lives of more than 6 million people worldwide (World Health Organization, 2020). The next section provides a brief overview of the placement of healthcare facilities in the context of the COVID-19 outbreak.

4.5 COVID-19 facility location

SARS-CoV-2 is responsible for causing COVID-19, which can range from a mild cold to Severe Acute Respiratory Syndrome requiring intensive hospital care (World Health Organization, 2020). The virus has rapidly spread worldwide and placed an extreme strain on health services, jeopardizing the quality of service. Politicians, decision-makers, researchers, health workers, and health service managers have joined efforts to face this health emergency.

An expert team evaluated 20 critical challenges in supply chain management in this context. According to the evaluation, the most significant challenge was found to be directly related to facility location (Karuppiyah et al., 2021). Although this is based on Indian expert feedback regarding the location of the COVID-19 facility, it is challenging for health services managers. Thus, further research is necessary to address any gaps in knowledge, as mentioned earlier.

The COVID-19 context presents a critical challenge due to the diverse range of services required, including prevention, basic and intensive care, and testing, which are complex and varied. To address this challenge, researchers have developed several solutions, such as identifying optimal locations for vaccination centers (Cabezas et al., 2021; Leithäuser et al., 2021; Çakır et al., 2021; Cabanilla et al., 2022; Goentzel et al., 2022), hospital beds (Delgado et al., 2022), location and allocation of material distribution (Wang & Ma, 2021), locating field hospitals (Hassan et al., 2021), establishing temporary testing laboratories (Devi et al., 2022) and mobile units (Vicencio-Medina et al., 2023). These solutions often employ multi-objective (Leithäuser et al., 2021; Lusiantoro et al., 2022; Delgado et al., 2022), dynamic (Cabanilla et al., 2022), gradual coverage (Delgado et al., 2022) approaches to tackle the problem effectively.

Actual data from India, Turkey, Egypt, Germany, Indonesia, Filipins, Gambia and China are used to test the models, which limits their applicability to other contexts (Hassan et al., 2021). On the contrary, Cabezas et al. (2021) employed simulated and benchmark instances, resulting in more generalizable models that performed well across different scenarios. The prevalence of real data in many articles indicates a desire to develop models that closely reflect actual situations and offer practical solutions.

The problems addressed are relatively minor, occurring only a limited number of times (no more than 150) and in a small number of installations (a maximal of 58). Therefore, the solution methods used were exact using solvers CPLEX, XPRESS, and GUROBI. However, other studies (Hassan et al., 2021; Wang & Ma, 2021; Çakır et al., 2021) have suggested heuristic and metaheuristic approaches to solve these problems, which have been compared to solver solutions and shown to provide results close to optimal values.

In conclusion, the findings demonstrate the efficacy of various methodologies, models, algorithms and heuristics / metaheuristics to obtain near-optimal solutions for various facility location issues in the context of COVID-19.

Despite this observation, the authors stressed the need for more research from a more practical standpoint. This includes exploring multi-facility, multimodal transportation, and multi-materials scenarios; utilizing online data; and extending testing and application to other regions and cities.

5 CONCLUSIONS

Choosing a suitable location for facilities is a complex problem that requires the attention of managers, researchers, and stakeholders from both the public and private sectors. This decision requires significant investments and has profound impacts on the quality of the services provided by these facilities. This core decision generally influences the supply chain, customer service, and even obtaining qualified labor.

This decision is even more critical for the health system, as it deals with people's lives. When this system faces an abrupt increase in demand, deciding where to locate facilities can increase the chances of saving lives, as experienced during the COVID-19 pandemic. This study conducted a systematic review of healthcare facility locations that focuses on maximal covering and exploration of the issue of hierarchical location and the context of COVID-19, covering a total of 203 articles since the first Church & ReVelle (1976).

The paper for hierarchical healthcare location focuses on two or three tiers, although in reality, health systems are usually three levels, with basic services, specialized services / hospitalizations and intensive care. Despite that, both approaches seek to model a more real problem as possible, encompassing uncertainties, capacitated systems, and others. The research opportunities in hierarchical location in health care are very broad, given the small number of articles in the literature, including:

- Consider service capacity;
- Adding demand and supply uncertainties to the model;
- Hierarchical approaches with more than two levels;
- Analyze patient facility selection behavior;
- Explore the use of stochastic programming and heuristics to solve large-scale problems;
- Using more realistic data to produce better solutions.

These features increase the complexity of the problem, making it impossible to use solvers such as CPLEX, GUROBI, and XPRESS. Therefore, near-optimal solutions can be obtained using heuristics and metaheuristics. On the other hand, solvers have been useful for smaller instances with few nodes and installations, or for comparing the results of heuristics.

In the context of COVID-19, research was even more limited to specific contexts, hence the highlighted opportunities.

- Developing and testing new models, e.g. using incremental covering;
- Using heuristics to solve strategies to reduce computational cost;
- Use timely updated data to make models more realistic;
- Test the models in other regions and countries to validate their applicability to other contexts.

When analyzing the general context of maximal covering in health care, we can add other possibilities:

- Use multi-objective models to bring the problem closer to reality;
- Develop efficient algorithms for use in large-scale problems, such as VNS (Variable Neighborhood Search) and tabu search, which have been used;
- Using real-world information about hospitals, disease spread patterns, census information, and distance between cities;
- Considering the issue of roads in rural areas.

Therefore, one of the greatest challenges in researching the location of healthcare services is to make the model as real as possible, even using updated, timely data if possible.

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The authors declare that there are no conflicts of interest with respect to the publication of this article.

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Do not apply.

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